Requirement Analysis

Hotel Booking Data Analysis and Business Intelligence Dashboard
Power BI Project 8 | Stage 1 of 7

Item	Detail
Project title	Hotel Booking Data Analysis and Business Intelligence Dashboard
Domain	Hospitality / hotel revenue management
Tool	Microsoft Power BI Desktop (Power Query, DAX, Report view)
Key concepts	Occupancy, cancellations, revenue
Data	Booking data for 2 hotels, Jan 2022 to Dec 2024, supplied with this project (see Dataset Guide)
Currency	Indian Rupees (INR)

1. Problem Statement

A hotel earns money only when its rooms are occupied at a good rate. About one booking in three is cancelled, often far in advance but sometimes close to arrival, which leaves rooms empty and makes demand hard to forecast. Management currently has raw booking records but no single view of how many rooms are sold, how many bookings are cancelled and why, which customers and channels bring the most revenue, and how performance changes through the year. This project builds a Power BI dashboard that answers these questions so that pricing, marketing and overbooking decisions are based on data.

2. Objectives

1. Clean and model the booking data into an analysis-ready star schema.
2. Measure overall performance: bookings, revenue, ADR (average daily rate), length of stay and occupancy.
3. Quantify cancellations and identify which customer groups, channels and booking behaviours drive them.
4. Identify seasonality, peak and off-peak periods, and year-on-year growth.
5. Identify the best-performing source countries, market segments and distribution channels.
6. Deliver an interactive three-page dashboard and a written set of business insights and recommendations.

3. Stakeholders and Needs

Stakeholder	Needs from the dashboard
General Manager	Overall revenue, occupancy and cancellation rate for each hotel; year-on-year growth.
Revenue Manager	ADR trends, seasonal demand, revenue lost to cancellations, effect of deposit policy.
Marketing Team	Top source countries, most valuable channels and segments, repeat guest share.
Front Office / Operations	Busy periods, average stay length, guest counts, room-type demand.

4. Business Questions

#	Business question	Area
Q1	How many bookings and how much revenue did each hotel generate each year?	Overview
Q2	What is the occupancy percentage of each hotel by month and year?	Occupancy
Q3	What percentage of bookings are cancelled, and how does it differ between the City Hotel and the Resort?	Cancellation
Q4	Does a longer lead time lead to more cancellations?	Cancellation
Q5	Which market segments and distribution channels bring the most revenue and the most cancellations?	Channels
Q6	How does deposit type (No Deposit, Refundable, Non Refund) affect cancellations?	Cancellation
Q7	Which months and seasons are busiest and which are slowest?	Seasonality
Q8	Which countries send the most guests and generate the most revenue?	Geography
Q9	How much revenue is lost to cancelled bookings?	Revenue
Q10	How do ADR and average length of stay change over time and by room type?	Revenue
Q11	What share of bookings come from repeat guests, and do they cancel less?	Customers

5. KPI Definitions

KPI	Definition	Source fields
Total Bookings	Count of booking records	booking_id
Cancelled Bookings	Bookings where is_canceled = 1	is_canceled
Cancellation Rate %	Cancelled bookings / total bookings	is_canceled
Total Revenue (INR)	ADR x total nights, for non-cancelled bookings	adr_inr, stays_in_weekend_nights, stays_in_week_nights, is_canceled
Average ADR	Average daily rate of confirmed bookings	adr_inr
Room-Nights Sold	Total nights of confirmed bookings	stays_in_weekend_nights, stays_in_week_nights
Occupancy %	Room-nights sold / (total rooms x days in period)	dim_hotel[total_rooms], arrival_date

KPI	Definition	Source fields
RevPAR (INR)	Total revenue / available room-nights	As above
Avg Lead Time	Average days from booking to arrival	lead_time
Avg Stay (Nights)	Average of total nights per booking	stays_in_weekend_nights, stays_in_week_nights
Revenue Lost (INR)	ADR x nights for cancelled bookings	adr_inr, nights, is_canceled
Repeat Guest %	Bookings with is_repeated_guest = 1 / total	is_repeated_guest

Occupancy note: the supplied dataset includes a hotel table with each hotel's total number of rooms (City Hotel 60, Resort 40). This allows a true occupancy percentage and RevPAR to be calculated, which most public booking datasets cannot do.

6. Data Requirements and Traceability

Each business question was mapped to the data needed to answer it. The supplied dataset contains every field listed here.

Question	Fields needed	Table
Q1, Q9	booking_id, hotel, arrival_date, adr_inr, nights, is_canceled	hotel_bookings
Q2	total_rooms, arrival and departure dates, is_canceled	dim_hotel + hotel_bookings
Q3, Q4	is_canceled, lead_time, hotel	hotel_bookings
Q5	market_segment, distribution_channel, adr_inr, is_canceled	hotel_bookings
Q6	deposit_type, is_canceled	hotel_bookings
Q7	arrival_date (year, month), bookings, nights	hotel_bookings + date table
Q8	country, adr_inr, nights	hotel_bookings + dim_country
Q10	adr_inr, room_type_reserved, nights	hotel_bookings + dim_room_type
Q11	is_repeated_guest, is_canceled	hotel_bookings

7. Dashboard Plan

Page	Purpose	Visuals
1. Overview	Headline performance	Cards: Total Bookings, Total Revenue, Cancellation %, Avg ADR, Occupancy %. Line: revenue and occupancy by month. Donut: confirmed vs cancelled. Bar: revenue by hotel. Slicers: Year, Hotel, Month.
2. Cancellations and Customers	Why bookings are lost	Bar: cancellation % by lead-time bucket, market segment, deposit type. Treemap: bookings by distribution channel. Column: repeat vs new guests. Card: revenue lost.

Page	Purpose	Visuals
3. Geography and Revenue	Where money comes from	Map: bookings by country. Bar: top countries by revenue. Treemap: revenue by segment. Matrix: month by hotel revenue.

8. Scope, Assumptions and Constraints

In scope: historical analysis of bookings for two hotels; one Power BI report with three pages; written insights.

Out of scope: real-time data, machine-learning forecasting, personal guest information.

Assumptions:

- The dataset is a realistic simulated dataset created for this project; patterns in it illustrate real hotel behaviour but are not a real company's results.
- Revenue = ADR x nights, counted only for non-cancelled bookings (cancelled and no-show bookings earn nothing).
- Bookings with zero guests, negative ADR and duplicate records are errors and will be removed in the cleaning stage.
- Extreme ADR values are treated as outliers and reviewed or removed in cleaning.
- Occupancy uses room-nights of confirmed bookings against the fixed room count in dim_hotel.

9. Deliverables

Deliverable	Description
Power BI file (.pbix)	Cleaned data, star-schema model, DAX measures, three-page dashboard
Project report	Problem, data, cleaning steps, model diagram, DAX measures, screenshots
Business insights	5 to 6 findings with recommendations

10. Project Plan and Success Criteria

Stage	Task	Status
1	Requirement Analysis (this document)	Complete
2	Dataset Collection (dataset supplied)	Complete
3	Data Cleaning in Power Query	Next
4	Data Modeling (star schema)	Pending
5	DAX measures and KPIs	Pending
6	Dashboard	Pending
7	Business Insights	Pending

The project is successful when: all 11 business questions can be answered from the dashboard; KPI values match a manual check on a filtered sample; slicers filter every visual correctly; and the

insights page gives clear, evidence-based recommendations.